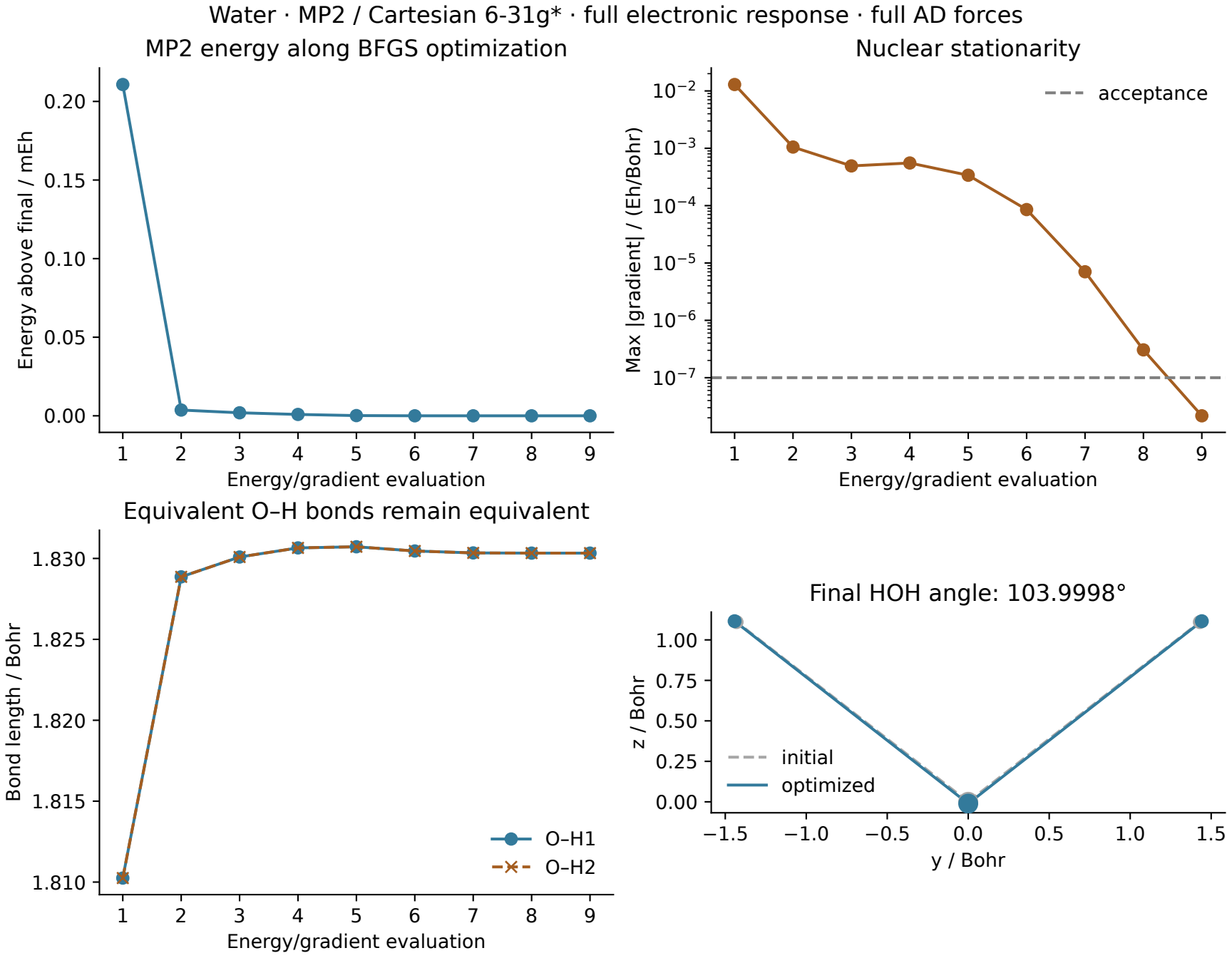
Water · MP2 / Cartesian 6-31g* · full electronic response · full AD forces
MP2 energy along BFGS optimization
Energy above final / mEh
0.20
0.15
0.10
0.05
0.00
1
2
3
4
5
6
7
8
9
Energy/gradient evaluation
Nuclear stationarity
Max |gradient| / (Eh/Bohr)
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
acceptance
Energy/gradient evaluation
Equivalent O–H bonds remain equivalent
Bond length / Bohr
1.830
1.825
1.820
1.815
1.810
O–H1
O–H2
Energy/gradient evaluation
Final HOH angle: 103.9998°
z / Bohr
1.00
0.75
0.50
0.25
0.00
initial
optimized
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
y / Bohr